

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12410827
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Fukuda; Naohiko et al.

Joint structure, method for designing joint structure, and method for manufacturing joint structure

Abstract

A joint structure increases fastening force between a first member and a second member while preventing peeling of an anti-rust coating film of the first member. The joint structure includes the first member in which the anti-rust coating film is formed by electrodeposition on a surface of a metallic base material; the second member that contacts the first member; and fastening members that fasten the first and second members together. Plural convex portions, each having a height less than a thickness of the electrodeposition coating film, are disposed in at least a part of a surface of the second member that contacts the first member. The first member and the second member are fastened by the fastening members such that the electrodeposition coating film of the first member and the convex portions mesh with each other without the base material of the first member contacting the convex portions.

Inventors: Fukuda; Naohiko (Aki-gun, JP), Kumagai; Takahiro (Aki-gun, JP), Miyamoto; Koji (Aki-gun, JP), Kimura; Hiroki (Aki-gun, JP)

Applicant: MAZDA MOTOR CORPORATION (Hiroshima, JP)

Family ID: 86227645

Assignee: MAZDA MOTOR CORPORATION (Hiroshima, JP)

Appl. No.: 18/051336

Filed: October 31, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230160408 A1	May. 25, 2023

Foreign Application Priority Data

JP	2021-191103	Nov. 25, 2021
----	-------------	---------------

Publication Classification

Int. Cl.: F16B5/02 (20060101)

U.S. Cl.:

CPC F16B5/02 (20130101);

Field of Classification Search

CPC: F16B (5/02); F16B (2/005); F16B (2200/506); F16B (33/008); F16B (37/00); F16B (39/282); F16B (5/00); F16B (5/065); B60G (13/003)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7014386	12/2005	Mullet	411/188	F16B 25/0021
2008/0247817	12/2007	Geislinger et al.	N/A	N/A
2012/0155988	12/2011	Schumacher	411/369	F16B 31/02
2017/0335869	12/2016	Bui et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10 2012 220 348	12/2013	DE	N/A
102015202037	12/2015	DE	F16B 2/005
102020105883	12/2020	DE	N/A
H01-266309	12/1988	JP	N/A

OTHER PUBLICATIONS

English translation DE102020105883 (Year: 2021). cited by examiner

English translation DE102015202037 (Year: 2016). cited by examiner

Office Action issued in DE 10 2022 129 537.0 ; mailed by the German Patent and Trademark Office on Jan. 8, 2025. cited by applicant

Primary Examiner: Vaughan; Jason L

Assistant Examiner: Kreiling; Amanda

Attorney, Agent or Firm: Studebaker Brackett PLLC

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a joint structure that fastens two members, a method for

designing the same, and a method for manufacturing the same.

BACKGROUND

(2) Conventionally, in a structure for fastening two members by a fastening member, in order to increase a fastening force it is known to subject a seat surface of the fastening member to surface treatment to increase a friction coefficient.

(3) For example, in a fastening structure disclosed in Japanese Patent document JP-A-1-266309, a ceramic plasma spray treatment is performed on a joint surface around a bolt hole to make a front surface of the joint surface around the bolt hole a rough surface. In this way, when a bolt is fastened to the bolt hole, a fastening force is increased by increasing a friction coefficient between a bolt head and the joint surface.

SUMMARY

(4) When a member to be fastened requires an anti-rust property, an electrodeposition coating such as cationic coating may be applied to a surface of a base material that constitutes the member. In a structure for fastening the member formed with an electrodeposition coating film on the base material surface to another member, in order to increase the friction coefficient, a washer is interposed between the bolt head and the member formed with the electrodeposition coating film, and a surface of the washer facing the electrodeposition coating film side is roughened as described above to form a convex portion.

(5) However, since an anti-rust coating film such as the electrodeposition coating film is extremely thin (about several tens of microns), the convex portion formed on the washer may penetrate the coating film and contact the base material surface of the member. In this case, a friction coefficient between the convex portion and the base material surface becomes a friction force factor, which worsens an effect of increasing the fastening force.

(6) In addition, when the convex portion penetrates the coating film and contacts the base material surface, a tip of the convex portion is likely to slide along the base material surface, which possibly peels off the coating film and forms rust on the base material surface.

(7) The present disclosure has been made in view of the above circumstance and therefore has a purpose of providing a joint structure in which a first member having an anti-rust coating film is joined to a second member, the joint structure capable of increasing a fastening force between the first member and the second member while preventing peeling of the anti-rust coating film of the first member.

(8) A joint structure according to the present disclosure includes a first member in which an anti-rust coating film is formed on a surface of a metallic base material; a second member that contacts the first member; and a fastening member that fastens the first member and the second member. Plural convex portions, a height of each of which is less than a thickness of the anti-rust coating film, are formed in at least a part of a surface of the second member that contacts the first member, and the first member and the second member are fastened by the fastening member such that the anti-rust coating film of the first member and the convex portion mesh with each other without the base material of the first member contacting the convex portion.

(9) In order to simultaneously exert two effects, which seem to be incompatible, of increasing a fastening force and maintaining anti-rust performance by the anti-rust coating film, the present disclosure has been completed by studying the height and a fastened state of the convex portion to increase a friction coefficient.

(10) According to the above configuration, the height of the convex portion of the second member is less than the thickness of the anti-rust coating film of the first member. Then, when the anti-rust coating film and the convex portion mesh with each other without the base material of the first member contacting the convex portion, the first member and the second member are fastened by the fastening member.

(11) In this joint structure, the anti-rust coating film and the convex portion mesh with each other without the base material of the first member contacting the convex portion. Accordingly, when the

second member is applied with a force in a sliding direction along the surface of the first member, in a state of maintaining a joined state between the base material of the first member and the anti-rust coating film, a force in a shearing direction is transmitted from the convex portion to the anti-rust coating film. Thus, due to shearing strength between the anti-rust coating film and the base material, it is possible to prevent the fastening member from sliding and to increase the fastening force between the first member and the second member.

(12) In addition, since the convex portion does not contact the base material, the anti-rust coating film is less likely to be peeled by relative displacement between the convex portion and the base material, and it is thus possible to maintain anti-rust performance by the anti-rust coating film.

(13) In the above joint structure, preferably, the convex portion has a sharply pointed tip.

(14) With such a configuration, the convex portion can reliably mesh with the anti-rust coating film.

(15) In the above joint structure, preferably, the second member is a plate material that overlaps the first member and is fastened.

(16) With such a configuration, the convex portion is formed on the plate material that overlaps the first member and is fastened. Accordingly, with a simple structure, it is possible to obtain a joint structure capable of increasing the fastening force while preventing peeling of the coating film on the first member. In addition, the convex portion is formed on the plate material as a fastening target of the first member, and thus, a dedicated member for forming the convex portion is no longer required in addition to the plate material. Therefore, it is possible to reduce weight of the joint structure and simplify the joint structure.

(17) In the above joint structure, preferably, the second member is a spacer, the spacer is interposed between a third member and the first member, and is fastened together with the first member and the third member, and the third member differs from the first member and the second member.

(18) With such a configuration, the convex portion is formed on the spacer that is interposed between the first member and the third member. Thus, with the simple structure, it is possible to obtain the joint structure capable of increasing the fastening force while preventing peeling of the coating film of the first member. In addition, a contact area between the spacer and the first member can be made smaller than an area of an opposing surface between the first member and the third member, which is advantageous for an increase in a friction force.

(19) In the above joint structure, preferably, the second member is a washer that is interposed between the first member and the fastening member.

(20) With such a configuration, the convex portion is formed on the washer that is interposed between the first member and the fastening member. Thus, with the simple structure, it is possible to obtain the joint structure capable of increasing the fastening force while preventing peeling of the coating film of the first member.

(21) A joint structure according to an embodiment of the present disclosure includes a first member in which an anti-rust coating film is formed on a surface of a metallic base material; a second member that overlaps the first member; and a fastening member that fastens the first member and the second member. The fastening member includes a stationary portion that does not rotate during fastening and that includes a surface contacting the anti-rust coating film of the first member. Plural convex portions, a height of each of which is less than a thickness of the anti-rust coating film, are formed in at least a part of a surface of the stationary portion. The first member and the second member are fastened by the fastening member such that the anti-rust coating film of the first member and the convex portion mesh with each other without the base material of the first member contacting the convex portion.

(22) With such a configuration, the plural convex portions, the height of each of which is less than the thickness of the anti-rust coating film, are formed on the surface, which contacts the anti-rust coating film of the first member, in the stationary portion, which does not rotate during fastening, of the fastening member, and the height of each of the convex portions is configured to be less than

the thickness of the anti-rust coating film. Then, in the state where the anti-rust coating film and the convex portion mesh with each other without the base material of the first member contacting the convex portion, the first member and the second member are fastened by the fastening member. In this joint structure, the anti-rust coating film and the convex portion mesh with each other without the base material contacting the convex portion. Thus, when any of the stationary portion of the fastening member and the first member is applied with the force in the sliding direction along the surface of the first member, the force in the shearing direction is input from the convex portion to the anti-rust coating film in a state of maintaining a joined state between the base material and the anti-rust coating film in the first member. Thus, due to shearing strength between the anti-rust coating film and the base material, it is possible to prevent the fastening member from sliding and to increase the fastening force between the first member and the second member. In addition, since the convex portion does not contact the base material, the anti-rust coating film is less likely to be peeled by relative displacement between the convex portion and the base material, and it is thus possible to maintain anti-rust performance by the anti-rust coating film.

(23) In the above joint structure, preferably, the joint structure is applied to fasten a suspension crossmember and a suspension in a vehicle, one of the suspension crossmember and the suspension includes the first member, and the other thereof includes the second member.

(24) With such a configuration, the above joint structure, in which the anti-rust coating film and the convex portion mesh with each other without the base material contacting the convex portion, is applied to fasten the suspension crossmember and the suspension, and the fastened portion therebetween is a portion applied with the large shearing force in the vehicle. In this way, it is possible to increase the fastening force without increasing a shaft diameter or an outer diameter of the fastening member. In addition, it is possible to reduce the weight of the fastening structure between the suspension crossmember and the suspension.

(25) In the above joint structure, preferably, the joint structure is applied to fasten a vehicle body and a suspension in a vehicle, one of the vehicle body and the suspension includes the first member, and the other thereof includes the second member.

(26) With such a configuration, the above joint structure, in which the anti-rust coating film and the convex portion mesh with each other without the base material contacting the convex portion, is applied to fasten the vehicle body and the suspension, and the fastened portion therebetween is a portion applied with the large shearing force in the vehicle. In this way, it is possible to increase the fastening force without increasing the shaft diameter or the outer diameter of the fastening member. In addition, it is possible to reduce the weight of the fastening structure between the vehicle body and the suspension.

(27) In the above joint structure, preferably, the joint structure is applied to fasten a battery unit, which is mounted on a vehicle, and a vehicle body, one of the battery unit and the vehicle body includes the first member, and the other thereof includes the second member.

(28) With such a configuration, the above joint structure, in which the anti-rust coating film and the convex portion mesh with each other without the base material contacting the convex portion, is applied to fasten the battery unit, which is mounted on the vehicle, and the vehicle body, and the fastened portion therebetween is a portion applied with the large shearing force in the vehicle. In this way, it is possible to increase the fastening force without increasing the shaft diameter or the outer diameter of the fastening member. In addition, it is possible to reduce the weight of the fastening structure between the battery unit and the vehicle body.

(29) A method for designing a joint structure according to the present disclosure in which a first member and a second member are fastened by a fastening member, the first member being formed with an anti-rust coating film on a surface of a metallic base material and the second member being formed with a convex portion in at least a part of a surface contacting the first member, includes a convex portion height determination step for determining a height of the convex portion to be less than a thickness of the anti-rust coating film; and an axial force determination step of determining

an axial force of the fastening member, with which the convex portion having the height determined in the convex portion height determination step can mesh with a coating film of the first member without the convex portion contacting the base material of the first member.

(30) In the above design method, in the convex portion height determination step, the height of the convex portion is determined to be less than the thickness of the anti-rust coating film. Then, in the axial force determination step, the axial force of the fastening member, with which the convex portion having the height determined in the convex portion height determination step can mesh with the coating film of the first member without the convex portion contacting the base material of the first member, is determined. In this design method, it is possible to reliably design the joint structure in which the convex portion can mesh with the anti-rust coating film without contacting the base material of the first member. In this way, it is possible to easily and reliably design the joint structure capable of increasing the fastening force while preventing peeling of the coating film of the first member.

(31) A method for manufacturing a joint structure according to the present disclosure that joins a first member and a second member includes a preparation step for preparing the first member in which an anti-rust coating film is formed on a surface of a metallic base material, the second member having a convex portion in a height less than a thickness of the anti-rust coating film in at least a part of a surface contacting the first member, and a fastening member that fastens the first member and the second member; a positioning step for positioning the first member and the second member such that the convex portion opposes the anti-rust coating film; and a fastening step for fastening the first member and the second member by the fastening member with an axial force of the fastening member capable of meshing with the anti-rust coating film without the convex portion contacting the base material of the first member.

(32) In the above manufacturing method, by sequentially executing the preparation step, the positioning step, and the fastening step described above, it is possible to reliably manufacture the joint structure in which the convex portion meshes with the anti-rust coating film without contacting the base material of the first member. In this way, it is possible to easily and reliably manufacture the joint structure capable of increasing the fastening force while preventing peeling of the coating film of the first member.

(33) In the joint structure according to the present disclosure, the first member having the anti-rust coating film is fastened to the second member, and can increase the fastening force between the first member and the second member while preventing peeling of the anti-rust coating film of the first member.

(34) In the method for designing the joint structure according to the present disclosure, it is possible to reliably design the joint structure capable of increasing the fastening force between the first member and the second member while preventing peeling of the anti-rust coating film of the first member.

(35) According to the method for manufacturing the joint structure according to the present disclosure, it is possible to reliably manufacture the joint structure capable of increasing the fastening force between the first member and the second member while preventing peeling of the anti-rust coating film of the first member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional explanatory view illustrating a joint structure according to a first embodiment of the present disclosure, in which a first member having an electrodeposition coating film as an anti-rust coating film and a plate material as a second member are fastened.

(2) FIG. 2 is a cross-sectional explanatory view illustrating a state where a convex portion formed

on an upper surface of the plate material in FIG. 1 meshes with the electrodeposition coating film without contacting a base material of the first member.

(3) FIG. 3 is a cross-sectional explanatory view illustrating, as a comparative example of FIG. 2, a state where a convex portion of the second member penetrates the electrodeposition coating film of the first member and contacts the base material in the conventional joint structure.

(4) FIG. 4 is a graph illustrating that a sliding friction coefficient is improved over an entire range of a surface pressure in the joint structure of this embodiment when compared that in the current joint structure as a comparative example.

(5) FIG. 5 is a cross-sectional explanatory view illustrating a joint structure according to a second embodiment of the present disclosure, in which a first member having an electrodeposition coating film, a third member, and a spacer as a second member having a convex portion are fastened in a state where the spacer is interposed between the first member and the third member.

(6) FIG. 6 is a cross-sectional explanatory view illustrating a joint structure according to a third embodiment of the present disclosure in which a washer as a second member having a convex portion, a first member having an electrodeposition coating film, and a third member are fastened in a state where the washer is interposed between the first member and a bolt head.

(7) FIG. 7 is a cross-sectional explanatory view illustrating a state where the convex portion formed on a lower surface of the washer in FIG. 6 meshes with the electrodeposition coating film without contacting a base material of the first member.

(8) FIG. 8 is a cross-sectional explanatory view illustrating a joint structure according to a fourth embodiment of the present disclosure, in which a first member having an electrodeposition coating film and a second member are fastened in a state where a convex portion is formed on a lower surface of a head of a fixing bolt as a stationary portion of a fastening member and the head contacts the first member.

(9) FIG. 9 is a cross-sectional explanatory view illustrating a fastened portion between a vehicle suspension system and a vehicle body as an application example of the joint structure in the present disclosure.

(10) FIG. 10 is a perspective explanatory view illustrating the vehicle suspension system and the surroundings thereof as another application example of the joint structure in the present disclosure.

(11) FIG. 11 is a perspective explanatory view illustrating a vehicle engine mount and the surroundings thereof as further another application example of the joint structure in the present disclosure.

(12) FIG. 12 is a cross-sectional explanatory view illustrating a joint structure according to a comparative example, in which a first member having an electrodeposition coating film, a third member, and a spacer as a second member having a convex portion are fastened in a state where the spacer is interposed between the first member and the third member.

DETAILED DESCRIPTION

(13) A detailed description will hereinafter be made of an embodiment of the present disclosure with reference to the accompanying drawings.

First Embodiment

(14) As illustrated in FIG. 1, a joint structure in a first embodiment includes: a first member 1 that is formed with an electrodeposition coating film 6 as an example of an anti-rust coating film on a surface 5a (see FIG. 2) of a metallic base material 5; a second member 2 that is arranged in a manner to overlap the first member 1 from below and contacts the first member 1; and a bolt 3 and a nut 4 as fastening members that fasten the first member 1 and the second member 2. The anti-rust coating films in the present disclosure include the electrodeposition coating film, an Al—Zn frame composite coating film, and the like. In this embodiment, the electrodeposition coating film 6 is an example of the anti-rust coating film.

(15) The base material 5 of the first member 1 is formed of a metal material, such as steel, that is generally used for a machine part. The electrodeposition coating film 6 is formed to have a uniform

film thickness on the surface 5a of the base material 5 by applying electrodeposition coating such as cationic coating to the base material 5. A through hole 1a in which the bolt 3 can be inserted is formed in the first member 1. A shape of the first member 1 is not limited to a particular shape in the disclosure, but a plate member may be employed as the first member 1.

(16) In this embodiment, the second member 2 is a plate material that overlaps the first member 1 and is fastened. In the second member 2, a through hole 2a in which the bolt 3 can be inserted is formed at a position that corresponds to the through hole 1a of the first member 1. The plate materials that are adopted as the first member 1 and the second member 2 each mean a material in a thickness that can be fastened by the fastening member, and include various materials, such as a press material, an extruded material, a casting, and a forging, that are not limited by manufacturing methods.

(17) The second member 2 is formed of a material that can firmly be fastened to the first member 1, and is manufactured by the metal material such as steel. However, the second member 2 can also be manufactured by using a hard resin or the like.

(18) As illustrated in FIG. 2, in at least a part of a surface of the second member 2 that contacts the first member 1 (that is, an upper surface 2b in FIG. 1), plural convex portions 7, a height X2 of each of the convex portions 7 is less than a thickness X1 of the electrodeposition coating film 6 are formed. The convex portion 7 only needs to be formed on at least a part of the surface of the second member 2 that contacts the first member 1. For example, in the first embodiment, the convex portions 7 are formed on the entire upper surface 2b of the second member 2, but may only be formed around the through hole 2a on the upper surface 2b.

(19) Here, in a fastened state of the bolt 3 and the nut 4, the thickness X1 of the electrodeposition coating film 6 is equal to a distance between a root end 7b of the convex portion 7 and the surface 5a of the base material 5.

(20) The thickness X1 of the electrodeposition coating film 6 is about several tens of microns (for example, 10 to 90 μm) and is extremely thin. As a method for forming the convex portion 7 in a height that is less than the thickness X1 of this electrodeposition coating film 6 (for example, a height that is less than 20 μm), for example, a technique of micro-fabricating a surface of a substrate in a unit of several microns in manufacturing of semiconductors, for example, shot blasting using ultra-fine particles is adopted.

(21) The convex portion 7 is formed to have a sharply pointed tip 7a. Thus, the convex portion 7 can easily bite into the electrodeposition coating film 6. An angle of the tip 7a of the convex portion 7 is set to such an angle at which the convex portion 7 can easily bite into the electrodeposition coating film 6 and the tip 7a is not damaged.

(22) The height of each of the convex portions 7, the angle of the tip 7a of each of the convex portions 7, a formation range of the convex portions 7, and a degree of the convex portions 7 being dense (density) are set in comprehensive consideration of a material characteristic of the second member 2 having the convex portions 7, a material characteristic of the electrodeposition coating film 6, another condition, and the like in order to obtain a required fastening force in the joint structure.

(23) As illustrated in FIG. 1, the first member 1 and the second member 2 are fastened by threading the nut 4 into a tip of the bolt 3 in a state where a shaft 3b of the bolt 3 penetrates the through holes 1a, 2a.

(24) In this fastened state, as illustrated in FIG. 2, the first member 1 and the second member 2 are fastened by the bolt 3 and the nut 4 in a state where the electrodeposition coating film 6 of the first member 1 meshes with the convex portion 7 without the base material 5 of the first member 1 contacting the convex portion 7.

(25) When the bolt 3 and the nut 4 are fastened, a head 3a of the bolt 3 and the nut 4 fasten the first member 1 and the second member 2 from both of upper and lower sides, respectively, which generates an axial force on the shaft 3b of the bolt 3.

(26) This axial force is managed by fastening torque of the bolt 3 and the nut 4.

Characteristics of First Embodiment

(27) (1)

(28) As illustrated in FIG. 2, the joint structure of the first embodiment is configured that the convex portion 7 of the second member 2 has the height X2 that is less than the thickness X1 of the electrodeposition coating film 6 of the first member 1. In addition, in the state where the electrodeposition coating film 6 and the convex portion 7 mesh with each other without the base material 5 of the first member 1 contacting the convex portion 7, the first member

contacting the first member, and a fastening member having a shaft that fastens the first member and the second member by passing through the first and second through holes; a positioning step for positioning the first member and the second member such that the convex portion opposes the anti-rust coating film; and a fastening step for fastening the first member and the second member by the fastening member with an axial force of the fastening member for meshing with the anti-rust coating film without the convex portion contacting the base material of the first member.
